

LECTURE 10: MATH1061/7861
TOPICS: PROOF BY CONTRAPOSITION, RATIONAL
NUMBERS
DATE: TUESDAY MARCH 17, 2026

Learning objectives:

$$p \rightarrow q \equiv \sim q \rightarrow \sim p$$

- (1) Learn how to write a proof by contraposition.
- (2) Reinforce proof techniques (direct proof, proof by contradiction, and proof by contraposition).
- (3) Understand the definitions of rational and irrational numbers.

* Check your grades on blackboard.

Feedback on grade scope.

* Please avoid going to Quiet

Applied classes (Thursdays noon
Fridays 1:00 p.m.)

Q1: Prove by contraposition:

$$\forall n \in \mathbb{Z}, \text{ if } 3n^2 \text{ is odd, then } n \text{ is odd.}$$

First step: Write the contrapositive.

$$\forall n \in \mathbb{Z} \text{ if } n \text{ is even then } 3n^2 \text{ is even.}$$

Suppose n is even. Then $n = 2K$
for some $K \in \mathbb{Z}$. Therefore,

$$3n^2 = 3(2K)^2 = 12K^2 = 2(6K^2)$$

So $3n^2$ is also even.

Q2: Prove by contraposition:

$\forall r \in \mathbb{R}$, if r^2 is irrational then r is irrational.

The contrapositive is :

$\forall r \in \mathbb{R}$ if r is rational then r^2 is rational.

Suppose r is rational. This means

$$r = \frac{a}{b} \quad a, b \in \mathbb{Z}, \quad b \neq 0.$$

$$\Rightarrow r^2 = \frac{a^2}{b^2} \quad a^2, b^2 \in \mathbb{Z}, \quad b^2 \neq 0.$$

$\Rightarrow r^2$ is also a rational number.

Q3: Prove or disprove: Counterexample: $r = \sqrt{2}$

- The sum of any two irrational numbers is irrational.

$$s = -\sqrt{2}$$

$$\equiv \forall r, s \in \mathbb{R} - \mathbb{Q}, \quad r + s \in \mathbb{R} - \mathbb{Q}.$$

$$\equiv \forall r, s \in \mathbb{R}, \quad r, s \notin \mathbb{Q} \longrightarrow r + s \notin \mathbb{Q}$$

Sum

- The ~~sum~~ of a rational and an irrational is irrational.

$$\forall r \in \mathbb{Q}, \quad \forall s \in \mathbb{R} - \mathbb{Q}, \quad r + s \in \mathbb{R} - \mathbb{Q}$$

See also next page.
for a reformulation.

← Equivalent
to statement
in next page.

Q4: Prove or disprove:

$\forall r \in \mathbb{Q}$, if s is irrational then $r + s$ is irrational.

Contrapositive: $\forall r \in \mathbb{Q}$, if $r + s$ is rational then s is rational.

Suppose r and $r + s$ are rational.

Then $r = \frac{a}{b}$, $r + s = \frac{c}{d}$ $a, b, c, d \in \mathbb{Z}$
 $b, d \neq 0$.

Then $s = (r + s) - r$
 $= \frac{c}{d} - \frac{a}{b}$
 $= \frac{bc - ad}{bd}$ $\leftarrow$ rational number
because $bc - ad \in \mathbb{Z}$
 $bd \in \mathbb{Z} - \{0\}$.